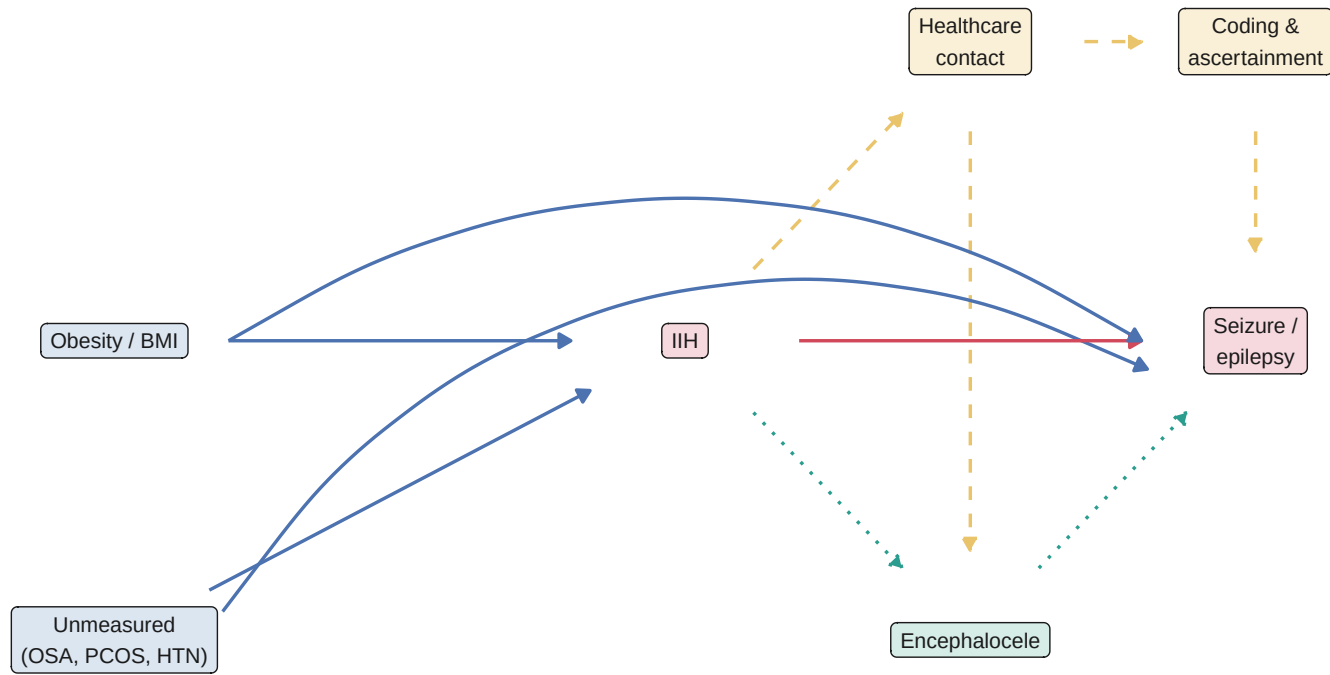


Assumed causal structure

Solid red = target effect. Dashed amber = the ascertainment paths the design must break.



→ CAUSAL EFFECT OF INTEREST → bias → confounding → mediation

Matching closes the obesity/age/sex path. The encephalocele mediation path (dotted) is SUSPENDED. Amber paths are addressed by the negative control, not by adjustment: conditioning on healthcare contact opens a collider path.